Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/12/2025
PI Name:	Juenger, Thomas Ph.D.		
Project Title:	Engineering Plant-Microbial Interactions for Improved Plant Growth on Simulated Lunar Regolith		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	01/01/2024	**End Date:**	12/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Jordan Risdal, a young technician, has left the project. Jordan is now a graduate student at at Baylor University in Houston Texas.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K0686		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants are remarkable for their ability to acquire nutrients and withstand abiotic stress across diverse soil substrates. This success is in part mediated by the interaction of roots with diverse microbial communities that facilitate the acquisition of nutrients, ameliorate various abiotic stresses, and protect plants from disease. However, the plant associated microbiome is remarkably diverse and little is known about the assembly, community dynamics, or drivers of the beneficial attributes of the plant microbiome. This complexity has hampered the engineering of beneficial microbiomes. Our aim is to engineer specific plant-microbiomes through successive bouts of selection that increase plant survival and growth on substrates of simulated lunar regolith. We will use an artificial selection strategy called host-mediated microbiome selection to engineer the microbiome and synthetic microbial community function. We will use sequencing to identify microbes associated with improved or decreased performance in the regolith selected lines. Bioinformatics analyses of the composition of taxa and gene content from the selected communities will identify enrichment of particular gene functions that improve tolerance to regolith. Ultimately, these studies will help to identify critical plant-microbe-soil traits that we can manipulate and engineer for improved plant growth on regolith.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plants are remarkable for their ability to acquire nutrients and withstand abiotic stress across diverse soil substrates. This success is in part mediated by the interaction of roots with diverse microbial communities. A future challenge for lunar space missions is the ability to grow plants on available lunar regolith as substrate. Our experiments will provide new insights into plant-microbial interactions, including providing a proof of concept for microbial selection experiments as tools for engineering plant performance.
Task Progress:	In year 2, we completed a highly controlled and replicated experiment with Brachypodium distachyon (a model C3 annual grass) and a microbial inoculum derived from a slurry of soil and rhizosphere of the desiccation-tolerant grass Tripogonella spicata, collected at the Enchanted Rock State Natural Area. Plants were grown with the addition of a separate treatment that includes soil derived from treated wastewater sludge (Dillo Dirt, City of Austin Waste Treatment, Austin, TX) as a natural carbon source to assess its impact on plant growth and microbial community composition. Here, we explored the impact of soil substrate and microbial inoculation on shoot biomass, tiller number, and tiller height. We chose to work with Brachypodium distachyon given its diminutive size, its close relationships to wheat, and our prior success selecting on stress tolerance microbiomes using it as a model system. One striking result from the experiment was complete failure of establishment of plants grown on profile and inoculated with microbes. The loss of replicates from this treatment combination suggests that an additional carbon source is necessary for early establishment of Brachypodium. We analyzed these data using a 2-way ANOVA and discovered a number of significant effects of inoculation and substrate on plant growth. Overall, we discovered that the combination of lunar highland regolith amended with Dillo dirt and microbial inoculation produced plants with the greatest shoot biomass and tiller number, although with modest tiller height. At harvest, roots were separated from shoots and root rhizosphere and soil samples will be collected for 16S sequencing to analyze the impact of substrate on microbial community composition. Ordination analyses reveal distinct microbial communities associated with the roots of plants grown in differing soil/inoculation contexts. Based on this initial experimental design and microbial inoculate, we initiated a series of co-propagation and selection cycles in an attempt to improve the performance of Brachypodium growth on the regolith + Dillo substrate. Our experimental design centered on ten independent selection lines, three random selection lines, and one sterile control. We allowed plants 35 days of growth and then harvested plants and the three largest plants were identified. Microbiomes from the selected individuals were pooled and we used this pool as microbial inoculant for the next round of selection. We collected glycerol stocks of the bulk microbiomes for every selection line for every round of selection. Overall, we completed seven cycles of selection over the past year. In general, we see a considerable boost in plant performance early in the selection regime that peaks at round 3. Subsequently, we see a substantial decline in performance of the selection lines at round 4. Despite ongoing selection, we see only minor recovery of the initial effect in the subsequent selection lines. We've also explored belowground patterns in the latter selection cycles, where we generally see a consistent suppression of root growth upon inoculation and no obvious improvement in response to selection targeted from the perspective of aboveground phenotypes. Evaluation of microbiol community composition of the first three selection cycles reveals a striking separation of the communities from the first 2 rounds of selection from the third. We are currently sequencing samples from the entire selection cycle, which should help us test alternative hypotheses concerning the pattern of selection response across the 7 cycles. We have three non-exclusive hypotheses to explain the pattern observed to date. First, the dramatic change in plant response between R3 and R4 may be due to a perturbation, either biotic or abiotic, that resulted in a dramatic shift in the microbial community composition. This shift resulted in a loss of the engineered benefits to the plant host. Second, we are concerned that the sterilization of dillo dirt for the establishment of R4 may have been incomplete, allowing for the invasion of the R3 selected inoculate by foreign (non-selected) microbiomes. Finally, it's possible that selection on aboveground biomass resulted in a tradeoff with belowground growth, resulting in a tipping point and failure at R4.
Bibliography Type:	Description: (Last Updated:)